**a)** ATC curve is also a 'U' shaped curve.

**b)** Initially the ATC declines, reaches a minimum when the plant is operated optimally, and rises beyond the optimum output.

**c)** The 'U' shape of the AC reflects the law of the variable proportions.

## 4.4.7 Marginal Cost (MC)

**Table 4.7** Marginal Cost

| Q (in unit) | TC (in ₹) | MC (in ₹) |
|---|---|---|
| 0 | 1000 | -- |
| 1 | 1200 | 1200 - 1000 = 200 |
| 2 | 1300 | 1300 - 1200 = 100 |
| 3 | 1400 | 1400 - 1300 = 100 |
| 4 | 1600 | 1600 - 1400 = 200 |
| 5 | 1900 | 1900- 1600 = 300 |

It is the cost of the last single unit produced. It is defined as the change in total costs resulting from producing one extra unit of output. In other words, it is the addition made to the total cost by producing one extra unit of output. Marginal cost is important for deciding whether any additional output can be produced or not. MC = ΔTC / ΔQ where MC denotes Marginal Cost, ΔTC denotes change in total cost and ΔQ denotes change in total quantity. For example, a firm produces 4 units of output and the Total cost is ₹ 1600. When the firm produces one more unit (4 +1 = 5 units) of output at the total cost of ₹ 1900, the marginal cost is ₹ 300.

MC = ₹ 1900 – ₹ 1600 = ₹ 300.

The other method of estimating MC is :

$$MC_n = TC_n - TC_{n-1}$$

where, 'MC' denotes Marginal Cost, '$TC_n$' denotes Total cost of 'n$^{th}$ item, $TC_{n-1}$ denotes Total Cost of 'n-1' $^{th}$ item. For example,

when $TC_4$ = ₹1600, $TC_{(4-1)}$=₹1400 and then MC= ₹200, (MC=1600-1400)

MC schedule is shown in Table 4.7 and MC Curve is shown in diagram 4.7.

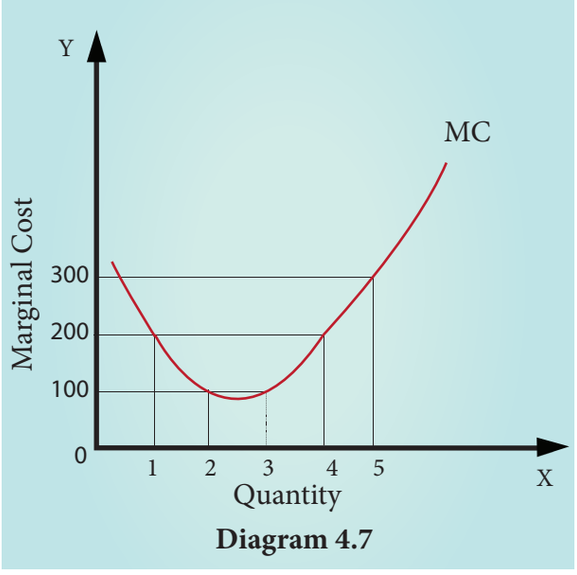


**Diagram 4.7**

It is to be noted that

**a)** MC falls at first due to more efficient use of variable factors.

**b)** MC curve increases after the lowest point and it slopes upward.

**c)** MC cure is a U-shaped curve.

**d)** The slope of TC is MC.

If TC = $Q^3 - 18Q^2 + 91Q + 12$

MC = $3Q^2 - 36Q + 91$



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